

Question				Marking details	Marks Available																				
					AO1	AO2	AO3	Total	Maths	Prac															
8	(a)	(i)		<table><tr><td></td><td>Actin</td><td>Myosin</td><td>Troponin</td><td>Tropomyosin</td></tr><tr><td>A Band</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>I Band</td><td>✓</td><td></td><td>✓</td><td>✓</td></tr></table> (1 mark per correct row)		Actin	Myosin	Troponin	Tropomyosin	A Band	✓	✓	✓	✓	I Band	✓		✓	✓	2			2		
	Actin	Myosin	Troponin	Tropomyosin																					
A Band	✓	✓	✓	✓																					
I Band	✓		✓	✓																					
		(ii)		Any 2 x (1) from: Myosin head binds to actin/cross bridges form between actin and myosin (1) Myosin head {bends/tilts/ power stroke} so actin is pulled past the myosin(1) Detail of ATP involvement in release of head (1)	2			2																	
	(b)	(i)		Calculate a mean to improve reliability / Identify that there were a range of sarcomere lengths (1)			1	1																	

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					AO1	AO2	AO3	Total	Maths	Prac
		(ii)		1.2 μm = 3 marks 1.2 (no units) = 2 marks (3.6-1.2) / 2 = 1 mark		3		3	3	
		(iii)		Any 2 x (1) The temperature of solution (1) pH of solution (1) same type of muscle (1) Same concentration solution (1) OR 2 controlled variables without solution (1)			2	2		2
		(iv)		Any 2 x (1) from: • no overlap between actin and myosin filaments (1) • no cross-bridges form / no possibility of myosin head moving actin (1) • no contraction possible / sarcomeres cannot get shorter (1)			2	2		
	(c)	(i)		each muscle contracts to 'pull' in each direction / as one contracts the other relaxes to move the limb (1)	1			1		
		(ii)		Any 2 x (1) from: Quadriceps contract (1) The osteoarthritic group has a lower contraction force ORA (1)	1	1		2		

		(iii)	Any 5 x (1) from: A. BMI states they are obese / heavier mass to be moved by quadriceps (1) B. Reference to BMI and the effect on contraction force/ stair climb (1) C. Exercise reduces BMI/ mass(1) D. Exercise strengthens the quadriceps (1) E. supports the joint/ strengthens muscles around the joint (1) F. increases flexibility of the joint / stimulates cartilage growth /reduces joint stiffness (1)		5		5		
			Question 8 total	6	9	5	20	3	2